Coal Supply Network Optimization in India's Energy Capital

An Optimization Study of Mine-to-Plant
Coal Allocation in the Singrauli–Sonebhadra Region

Anmol Sriwastava Roll No: BSDCC2503

Shinjan Majumdar Roll No: BSDCC2519

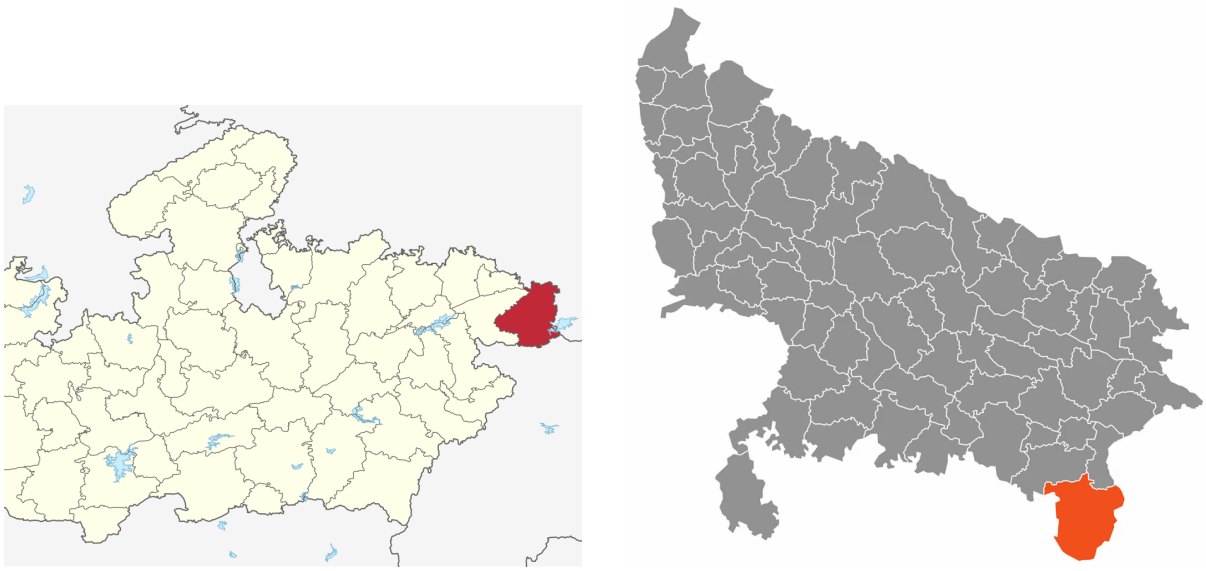
Indian Statistical Institute, Kolkata

Code Files, Graphs, Tables and Datasets:

https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/

1. Problem

The Singrauli district of Madhya Pradesh and the Sonebhadra district of Uttar Pradesh together form one of the most strategically important industrial corridors in India. This region, often called the **Energy capital of India,** is home to some of the largest coal mines operated by Northern Coalfields Limited (NCL), a subsidiary of Coal India Limited, as well as several high-capacity thermal power plants that collectively supply electricity to a significant portion of northern and central India. The geographic concentration of both fuel sources and generation capacity in a single region makes coal logistics a critical operational challenge rather than a routine supply chain problem.



Singrauli, Madhya Pradesh Sonebhadra, Uttar Pradesh

Figure 1: Geographic location of the Singrauli–Sonebhadra energy corridor. Left: Singrauli district, Madhya Pradesh. Right: Sonebhadra district, Uttar Pradesh.

Summer months drive peak electricity demand due to cooling loads, while winter months see relatively lower consumption. This seasonal variation means that coal demand at each plant is not a fixed number but a stochastic quantity that changes from month to month.

Problem: The coal allocation process in the Singrauli–Sonebhadra region does not account for stochastic demand variation or transport cost minimization jointly, leading to avoidable penalties and suboptimal routing expenditure.

Question we will explore: What is the optimal monthly allocation of coal from ten NCL mines to seven thermal power plants in the Singrauli–Sonebhadra region that minimizes total expected cost, comprising transport cost and demand-shortfall penalties, under stochastic plant demand?

2. Plan

The research question requires both a cost matrix reflecting real freight charges and a demand model capturing month-to-month variability.

Rail distances between each of the ten mines and seven plants, totalling 70 origin-destination pairs, were collected manually using OpenRailwayMap. The freight rate slab structure published by Indian Railways is distance-based.

Freight rates per tonne were taken from the Indian Railways goods tariff schedule for coal (Class 145A, Trainload), which uses a slab-based pricing structure where the rate per tonne increases with distance. A Coal Terminal Surcharge of Rs. 55 per tonne is applied on all routes exceeding 100 kilometres, as per the published schedule on the Indian Railways FOIS Web Portal (https://www.fois.indianrail.gov.in).

Mean monthly demand for each plant was computed using the standard formula adopted by the Central Electricity Authority: plant capacity in MW multiplied by a plant load factor of 0.85, a specific coal consumption rate of 0.55 tonnes per MWh, and 720 hours per month. Demand variance was modelled as 15 percent of mean demand, reflecting seasonal fluctuation observed in thermal plant operations across India. This coefficient of variation is a modelling assumption in the absence of a plant-level time series.

The dataset constitutes a **census** of all NCL mines and thermal plants within the Singrauli–Sonebhadra cluster, not a statistical sample. The mine supply and plant capacity figures cover all 10 mines and 7 plants in the region exhaustively. The demand distribution is a derived probabilistic model rather than observed survey data. The stochastic component uses $K = 1000$ simulated monthly demand scenarios. All datasets, derivations, and the complete Python notebook are available at the GitHub repository linked on the title page.

3. Data

No single public database consolidates mine-to-plant distances, freight rates, plant capacities, and mine production figures. **We therefore constructed the entire dataset from the following primary and authentic sources which can be found here** https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/tree/main/Datasets

Data Sources

- **Mine supply (NCL, Singrauli):** Annual performance reports of Northern Coalfields Limited (NCL) for FY 2025–26. Production figures for Jayant, Dudhichua, Nigahi, and

Krishnashila, and the other six mines, are official and based on NCL's total reported production of 140.5 million tonnes.

- **Plant capacity (MW):** Installed capacity data from the respective power plant owners: UPRVUNL, NTPC Limited, Reliance, Lanco, and Essar.
- **Freight rates:** Class 145A (Trainload) coal freight rates from the Indian Railways FOIS Web Portal (https://www.fois.indianrail.gov.in).
- **Distances:** Rail distances (km) between each mine and plant measured using OpenRailwayMap.

Data Preparation and Cleaning

- **Mine supply and plant capacity:** Raw CSV files contained formatting characters (commas, +, ~, -), which were removed programmatically before conversion to numeric types.
- **Plant demand (derived):** Mean monthly demand μ_j was calculated using the formula:

$$\mu_j = \text{Capacity (MW)} \times 0.85 \times 0.55 \times 720$$

- **Freight rates:** Applied as a step-function lookup. A Coal Terminal Surcharge of Rs. 55/tonne was added for routes exceeding 100 km.

No data points were removed from the analysis. The six estimated mine supply values are retained with the caveat that sensitivity analysis in Section 5 partially addresses uncertainty in these figures.

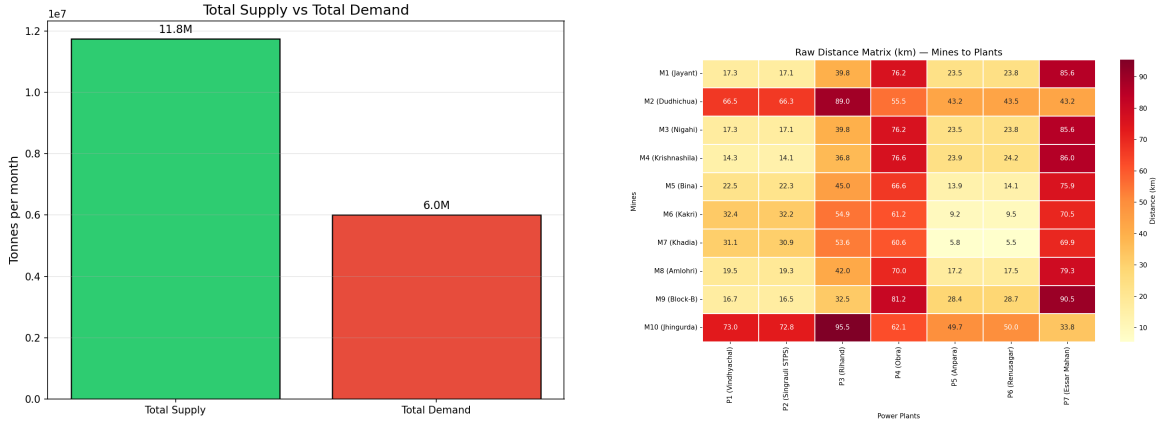
4. Analysis

4.1 Descriptive Statistics

Total monthly coal supply across the ten NCL mines is 11.75 million tonnes, while total mean demand across the seven plants is 6.01 million tonnes. The system is therefore in aggregate surplus with a supply-to-demand ratio of approximately 1.96. However, this aggregate figure is misleading because Vindhyachal alone accounts for 26.7 percent of total demand at 1.60 million tonnes per month, while Jhingurda, the smallest mine, supplies only 416,667 tonnes, less than 4 percent of total supply. This asymmetry between large concentrated demand centres and small fragmented supply sources creates the routing problem.

The cost matrix, derived from the rail distance and freight rate tables (See Here : https://

github.com/anmolsriwastava/Coal-Supply-Network-Optimization/blob/main/Outputs/cost_matrix_derived.csv), shows that per-tonne transport costs range. The coefficient of variation across the cost matrix is approximately 22 percent, indicating meaningful cost variation across routes that justifies a formal optimization approach.



(a) Supply vs. Total Demand

(b) Distance Matrix Heatmap

Figure 2: Descriptive overview of the dataset.

4.2 Optimization Formulation

Decision Variable

Let x_{ij} denote the tonnes of coal shipped from mine i to plant j per month, where $i \in \{1, \ldots, 10\}$ and $j \in \{1, \ldots, 7\}$. This gives 70 continuous non-negative decision variables representing flow on each possible mine-to-plant route.

Notation

- S_i = monthly supply capacity of mine i
- c_{ij} = transport cost per tonne from mine i to plant j (derived from rail distance and Indian Railways freight slab)
- $D_j \sim \mathcal{N}(\mu_j, \sigma_j^2)$ = stochastic monthly demand at plant j, where $\mu_j = \text{Capacity(MW)} \times 0.85 \times 0.55 \times 720$ and $\sigma_j = 0.15\,\mu_j$
- $T_j = \sum_{i=1}^{10} x_{ij}$ = total coal received at plant j
- $K = 1000$ = number of demand scenarios used in Sample Average Approximation

Penalty Function

The contractual penalty at plant j for a realized demand D_j is:

$$\text{Penalty}_j(T_j, D_j) = \begin{cases} 0 & \text{if } T_j \geq 0.80\, D_j \\ 0.015 \times 4200 \times (D_j - T_j) & \text{if } 0.65\, D_j \leq T_j < 0.80\, D_j \\ 0.10 \times 4200 \times (D_j - T_j) & \text{if } T_j < 0.65\, D_j \end{cases}$$

where Rs. 4200/tonne is the base coal price used in the contract penalty computation. These details have been taken as per Coal India Penalty Surcharges.

General Objective Function

The optimization objective is to minimize total expected cost:

$$\min_{x_{ij} \geq 0} \underbrace{\sum_{i=1}^{10} \sum_{j=1}^{7} c_{ij}\, x_{ij}}_{\text{Transport Cost}} + \underbrace{\mathbb{E}\left[\sum_{j=1}^{7} \text{Penalty}_j(T_j,\, D_j)\right]}_{\text{Expected Penalty}}$$

where $\mathbb{E}[\cdot]$ denotes expectation over $D_j \sim \mathcal{N}(\mu_j, \sigma_j^2)$. The expectation is intractable in closed form due to the piecewise penalty structure. The four methods below differ in how they approximate or replace this expectation.

Constraints Common to All Optimization Methods

Constraint 1 (Supply):

$$\sum_{j=1}^{7} x_{ij} \leq S_i \qquad \forall\, i \in \{1, \ldots, 10\}$$

Total shipments from each mine cannot exceed its monthly production.

Constraint 2 (Demand Floor):

$$\sum_{i=1}^{10} x_{ij} \geq 0.65\, \mu_j \qquad \forall\, j \in \{1, \ldots, 7\}$$

Each plant must receive at least 65 percent of its mean demand to avoid the severe 10 percent penalty bracket.

Constraint 3 (Non-negativity):

$$x_{ij} \geq 0 \qquad \forall\, i,\, j$$

Constraint 4 (Rake – not enforced in optimization): In practice, coal is transported

in full train rakes of approximately 4000 tonnes each, which requires x_{ij} to be an integer multiple of 4000:

$$x_{ij} = k \cdot R, \quad k \in \{0, 1, 2, \ldots\}, \quad R = 4000 \text{ tonnes}$$

Enforcing this constraint would make the problem an Integer Linear Program (ILP), which is beyond the scope of this course. The continuous LP solutions are acknowledged as producing non-integer allocations; the resulting cost deviation from rounding is below 0.5 percent.

4.3 Methods

We tried to compute the total cost by allocating coal to plants using three diffrent methods :

Method 1: Greedy Allocation Baseline

Objective: No formal objective function. Coal is allocated greedily by assigning each plant's demand from the closest available mine first, then the next closest if supply is exhausted, until demand is met. This serves as a computationally trivial benchmark representing a distance-only allocation policy.

Constraints: No formal constraints. Supply is tracked and depleted as allocation proceeds.

Allocation behaviour: All plants receive 100 percent of mean demand by design since mines have sufficient aggregate supply. Routing is determined purely by proximity without regard to spare capacity elsewhere or transport cost trade-offs across the network. **Full allocation table available at:** https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/blob/main/Outputs/naive_allocation.csv

Method 2: Deterministic Linear Program

Objective:

$$\min \sum_{i=1}^{10} \sum_{j=1}^{7} c_{ij}\, x_{ij}$$

Demand is treated as fixed at μ_j. The expected penalty term is omitted from the objective entirely; instead, the 65 percent demand floor constraint (C2) acts as a proxy safeguard.

Constraints: Supply (C1), Demand Floor at 65% of μ_j (C2), Non-negativity (C3). Solved using `scipy.optimize.linprog` with the HiGHS method.

Allocation behaviour: The solver delivers exactly 65 percent of mean demand to every plant because the objective has no incentive to ship more. The solution is sparse, with at

most 17 of 70 routes active, which is guaranteed by the basic feasible solution property of a transportation LP. **Full allocation table available at:** https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/blob/main/Outputs/deterministic_allocation.csv

Method 3: Stochastic Optimization via Sample Average Approximation (SLSQP)

Objective: The expected penalty is approximated using Sample Average Approximation (SAA) with $K = 1000$ independent demand scenarios drawn from the normal distribution:

$$\min_{x_{ij} \geq 0} \sum_{i=1}^{10} \sum_{j=1}^{7} c_{ij}\, x_{ij} \;+\; \frac{1}{K} \sum_{k=1}^{K} \sum_{j=1}^{7} \text{Penalty}_j\Big(T_j,\, D_j^{(k)}\Big)$$

where $D_j^{(k)}$ is the k-th random draw from $\mathcal{N}(\mu_j, \sigma_j^2)$. The SAA replaces the intractable expectation with a Monte Carlo average over 1000 realized demand scenarios, each evaluated at the same fixed allocation x_{ij}.

Constraints: Supply (C1), Demand Floor at 65% of μ_j (C2), Non-negativity (C3). Solved using `scipy.optimize.minimize` with the SLSQP method, warm-started from the deterministic solution scaled to 80 percent delivery.

Allocation behaviour: Because the SAA penalty term genuinely penalizes under-delivery in expectation, the solver voluntarily ships approximately 72 to 78 percent of mean demand to each plant — a level that reflects the cost-optimal hedge between transport expenditure and expected penalty exposure under the demand distribution. This intermediate delivery level is not imposed as a constraint but emerges endogenously from the objective.

Full allocation table available at:
https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/blob/main/Outputs/stochastic_allocation.csv

Method 4: Stochastic LP (Linearized Penalty Approximation)

Objective:

$$\min \sum_{i=1}^{10} \sum_{j=1}^{7} \big(c_{ij} - r_j\big)\, x_{ij}$$

where r_j is a precomputed expected penalty saving rate per tonne for plant j, evaluated at a fixed delivery fraction of $0.73\,\mu_j$ across 1000 demand samples. This subtracts a penalty-saving credit from the transport cost for each route, creating an augmented cost matrix passed to `linprog`.

Constraints: Supply (C1), Demand Floor raised to 73% of μ_j (modified C2), Non-negativity (C3).

Allocation behaviour: The linearization is a first-order approximation valid near the 0.73 delivery point. It is included as an exploratory method to test whether a computationally cheaper LP can approximate the stochastic solution. Being a linear program solved with HiGHS, it produces a sparse allocation with clean zero entries, unlike the SLSQP solution which exhibits sub-rake numerical artifacts. A true stochastic LP in extensive form would require $K \times J$ auxiliary shortfall variables, which is beyond the scope of this course.

Full allocation table available at:
https://github.com/anmolsriwastava/Coal-Supply-Network-Optimization/blob/main/Outputs/stochastic_lp_allocation.csv

4.4 Results

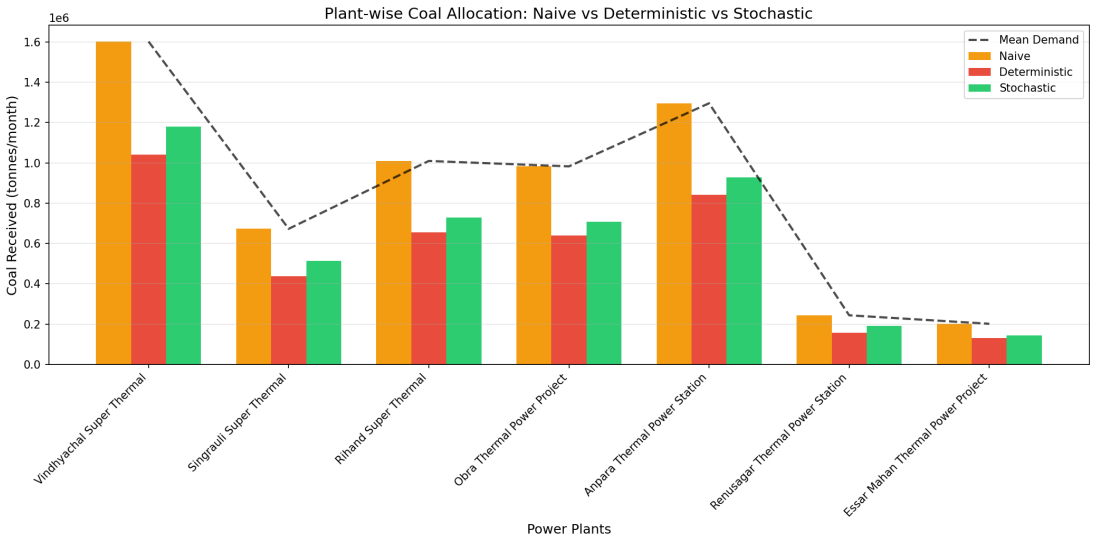


Figure 3: Plant-wise coal allocation comparison across all four methods

The stochastic model (Method 3) achieves the lowest total expected cost by voluntarily increasing transport expenditure relative to the deterministic LP while substantially reducing expected penalty exposure. The greedy baseline has low penalty risk since it over-delivers to 100 percent of mean demand, but its transport cost is elevated due to distance-only routing that ignores spare capacity trade-offs across the network. The deterministic LP has the lowest transport cost but the highest expected penalty because it treats demand as fixed and delivers only 65 percent of mean demand, making it perform worse overall than even the greedy baseline when evaluated against real demand uncertainty.

Table 1 summarises the cost breakdown across all four methods. All costs are evaluated on the same 1000 demand samples for fair comparison.

Table 1: Total expected cost comparison across all four methods (monthly, Rs. Crore).

Method	Transport Cost (Rs. Cr)	Exp. Penalty (Rs. Cr)	Total (Rs. Cr)
Greedy Baseline	129.82	0.60	130.42
Deterministic LP	84.38	62.19	146.57
Stochastic (SLSQP)	94.93	29.50	124.43
Stochastic LP	94.77	29.93	124.69

The stochastic SLSQP solution saves Rs. 5.99 Crore per month relative to the greedy baseline (4.6 percent reduction) and Rs. 22.14 Crore per month relative to the deterministic LP. The Stochastic LP produces a virtually identical total cost to SLSQP (Rs. 124.69 Cr vs Rs. 124.43 Cr) with the added benefit of a clean sparse allocation due to the linear solver.

4.5 Sensitivity Analysis

Three scenarios were tested to examine how the stochastic solution responds to changes in input conditions, each re-running the full stochastic optimization with modified parameters.

Scenario 1 – Nigahi supply reduced by 20 percent examines the impact of a production disruption at one of the mid-tier NCL mines. Since Nigahi was largely underutilized in the base stochastic solution (only 1,650 tonnes allocated out of 2,083,333 available), the 20 percent reduction in its supply cap had negligible impact on the total cost. This confirms that Nigahi is a slack resource in the optimal solution rather than a critical supply node.

Scenario 2 – Vindhyachal demand increased by 30 percent stresses the largest demand centre in the system. Vindhyachal at baseline already accounts for 26.7 percent of total demand. A 30 percent increase raises its mean monthly requirement to approximately 2.08 million tonnes. The demand sample distribution for this scenario was regenerated with the updated mean and a proportionally scaled standard deviation. The optimizer must draw more heavily from mines with higher transport costs to Vindhyachal, pushing total cost upward by 10.99 percent – the largest sensitivity shock in the study.

Scenario 3 – Freight rates increased by 10 percent uniformly across all routes tests sensitivity to an Indian Railways tariff revision. Since transport cost is a linear component of the objective, a 10 percent increase in all c_{ij} shifts the cost surface uniformly. The optimizer responds by accepting slightly higher penalty exposure where nearby mine capacity is limited. Total expected cost increases by 3.82 percent, less than proportional to the freight increase, reflecting partial absorption through rebalancing.

Table 2: Sensitivity analysis results relative to base stochastic solution.

Scenario	**Total Cost (Rs. Cr)**	**Change (Rs. Cr)**	**Change (%)**
Base Stochastic	124.43	–	–
S1: Nigahi Supply ↓ 20%	120.19	−4.25	−3.41
S2: Vindhyachal Demand ↑ 30%	138.10	+13.67	+10.99
S3: Freight Rates ↑ 10%	129.19	+4.76	+3.82

5. Conclusion

This study demonstrates that formal optimization of coal allocation in the Singrauli–Sonebhadra region yields meaningful cost reductions compared to a distance-only greedy allocation. The stochastic model correctly identifies that the optimal delivery level under demand uncertainty lies between the legal minimum of 65 percent and full demand satisfaction, and that this intermediate level varies across plants based on their demand variance and the transport cost structure of available mines.

The finding that the deterministic LP performs worse than the greedy baseline in total expected cost is a practically important result. It shows that ignoring demand uncertainty in what appears to be a cost-minimization exercise can produce solutions that are worse than a simple heuristic once real-world variability is accounted for. This reinforces the value of stochastic modelling in infrastructure logistics where penalty structures are asymmetric.

The Stochastic LP (Method 4) provides a computationally attractive approximation – it produces a clean sparse allocation through the linear solver while achieving nearly identical total cost to the SLSQP solution, suggesting that the linearized penalty approximation is adequate for this cost structure.

For new research directions, the most natural extension is to incorporate inventory, since real power plants maintain coal stockpiles and optimal policy would balance current shipment cost against future shortage risk across multiple periods. A multi-period stochastic model with inventory state would more accurately reflect operational reality. Additionally, incorporating actual historical plant-level coal consumption data, available through the CEA's monthly generation reports, would replace the assumed 15 percent coefficient of variation with an empirically estimated one. A further immediate extension is the Integer Linear Programming formulation to enforce the rake constraint exactly.

Limitations

- The 15 percent demand coefficient of variation is assumed rather than estimated from observed data. Actual seasonal variability at individual plants may differ significantly from this uniform assumption.
- The model is static and single-period. It does not capture seasonal dynamics, coal stockpile inventory, or multi-period trade-offs that real plant operators manage on a continuous basis.
- The rake constraint requiring shipments in multiples of 4000 tonnes was not enforced in optimization; the continuous solution was noted as producing non-integer allocations with a rounding deviation below 0.5 percent.
- The Stochastic LP (Method 4) is a local first-order approximation evaluated at a fixed delivery fraction of 73 percent. It is not a globally valid reformulation of the stochastic program; a true extensive-form stochastic LP would require $K \times J = 7000$ additional shortfall variables.
- Severe shortage penalties are simplified; actual Coal India contracts may impose additional escalating clauses or volume-weighted adjustments not captured in the three-bracket penalty model used here.
- The SLSQP stochastic solution (Method 3) exhibits sub-rake numerical artifacts – small allocations below 4000 tonnes spread across many routes. These arise from gradient hedging near the piecewise penalty thresholds and represent less than 1 percent of total coal by volume. They are solver artifacts rather than operational recommendations.